1 Solution — GIAM 1.2 Problem 4

1.1 Problem

Characterize the prime factorizations of numbers that are perfect squares.

1.2 The Characterization

Write a positive integer $n > 1$ in its prime factorization

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k},$$

with distinct primes $p_1 < p_2 < \cdots < p_k$ and exponents $a_i \geq 1$. Then:

$$\boxed{n \text{ is a perfect square} \iff \text{every exponent } a_i \text{ is even.}}$$

(And $n = 1 =$ the empty product is a perfect square, $1 = 1^2$, vacuously satisfying "every exponent even.")

1.3 Proof

Both directions lean on the **Fundamental Theorem of Arithmetic**: every positive integer has a prime factorization that is *unique* up to the order of the factors.

($\Leftarrow$) Even exponents $\Rightarrow$ perfect square. Suppose every a_i is even, say $a_i = 2b_i$ with $b_i \geq 0$ an integer. Set

$$m = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}.$$

Then

$$m^2 = \left(p_1^{b_1} \cdots p_k^{b_k}\right)^2 = p_1^{2b_1} \cdots p_k^{2b_k} = p_1^{a_1} \cdots p_k^{a_k} = n.$$

So $n = m^2$ is a perfect square.

($\Rightarrow$) Perfect square $\Rightarrow$ even exponents. Suppose $n = m^2$ for some positive integer m. Factor m by the FTA:

$$m = q_1^{c_1} q_2^{c_2} \cdots q_j^{c_j}.$$

Squaring,

$$n = m^2 = q_1^{2c_1} q_2^{2c_2} \cdots q_j^{2c_j}.$$

This is *a* prime factorization of n in which every exponent $(2c_i)$ is even. By the *uniqueness* half of the FTA, it is *the* prime factorization of n — so the primes q_i are exactly the p_i, and the exponents $a_i = 2c_i$ are all even. ■

The uniqueness step is essential: it is what lets us conclude that the *only* way to factor n has even exponents, rather than merely that *one* such factorization exists.

1.4 The Idea in One Picture

A perfect square is a number whose prime factors can be dealt out evenly into **two identical hands** — one copy of $\sqrt{n}$ for each. An even exponent on each prime is exactly the condition that lets every prime be split down the middle.

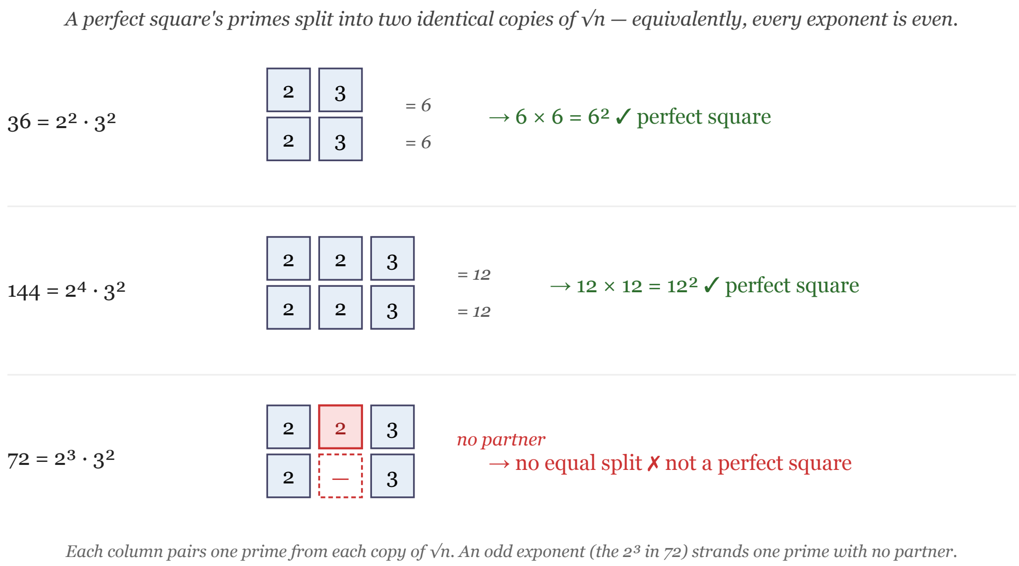

Figure 1.1: Top: $36 = 2^2 \cdot 3^2$ and $144 = 2^4 \cdot 3^2$ split into two equal copies (6×6 and 12×12). Bottom: $72 = 2^3 \cdot 3^2$ has an odd exponent on 2, so one factor of 2 is stranded with no partner — 72 is not a perfect square.

For $36 = 2^2 \cdot 3^2$ the primes split as $\{2, 3\} \mid \{2, 3\}$, giving 6×6. For $144 = 2^4 \cdot 3^2$ they split as $\{2, 2, 3\} \mid \{2, 2, 3\}$, giving 12×12. But $72 = 2^3 \cdot 3^2$ has an *odd* number of factors of 2: two of them pair up, the third has no partner, so 72 cannot be written as $m \times m$.

1.5 Examples

n	**Prime factorization**	**All exponents even?**	**Perfect square?**
36	$2^2 \cdot 3^2$	yes	yes, 6^2
144	$2^4 \cdot 3^2$	yes	yes, 12^2
225	$3^2 \cdot 5^2$	yes	yes, 15^2
400	$2^4 \cdot 5^2$	yes	yes, 20^2
72	$2^3 \cdot 3^2$	no (2^3)	no
50	$2 \cdot 5^2$	no (2^1)	no
105	$3 \cdot 5 \cdot 7$	no (all ^{1})	no (square-free)

Table 1.1

1.6 Remark — Equivalent Restatement via the Divisor Count

There is a second, equally clean characterization of perfect squares, in terms of the **number of divisors**. If $n = p_1^{a_1} \cdots p_k^{a_k}$, the count of positive divisors of n is

$$d(n) \;=\; (a_1 + 1)(a_2 + 1) \cdots (a_k + 1),$$

since a divisor is formed by choosing each exponent independently from 0 to a_i. This product is **odd** if and only if every factor $a_i + 1$ is odd, i.e. every a_i is even. Combining with our characterization:

$$n \text{ is a perfect square} \iff d(n) \text{ is odd.}$$

The intuitive reason: divisors of n pair up as $\{d,\, n/d\}$, and the pairing has no fixed point *unless* some $d = n/d$, i.e. $d = \sqrt{n}$. So the divisors come in pairs except when n is a square, in which case $\sqrt{n}$ is its own partner — leaving an *odd* total. This is why, for instance, 36 has the odd number of divisors $1, 2, 3, 4, 6, 9, 12, 18, 36$ (nine of them), with 6 as the unpaired middle.

1.7 Remark — Connection to Irrationality of $\sqrt{n}$

Our characterization is the engine behind the fact (used in Section 1.1, Problem 6) that $\sqrt{n}$ is irrational unless n is a perfect square. Writing $n = p_1^{a_1} \cdots p_k^{a_k}$:

$$\sqrt{n} \;=\; p_1^{a_1/2} \cdots p_k^{a_k/2}.$$

If every a_i is even, all the exponents $a_i/2$ are integers and $\sqrt{n}$ is the *integer* $p_1^{a_1/2} \cdots p_k^{a_k/2}$. But if even one exponent a_j is odd, then $a_j/2$ is a half-integer; a short argument (essentially the one for $\sqrt{2}, \sqrt{3}$ in Problem 6) shows no rational number can have such a factorization, so $\sqrt{n}$ is irrational. In short:

$$\sqrt{n} \in \mathbb{Q} \iff \sqrt{n} \in \mathbb{Z} \iff n \text{ is a perfect square.}$$

For square roots there is no middle ground: $\sqrt{n}$ is either a whole number or irrational — never a non-integer rational.

1.8 Remark — Generalization to Higher Powers

The characterization upgrades verbatim to **perfect k-th powers**:

$$n \text{ is a perfect } k\text{-th power} \iff k \mid a_i \text{ for every } i,$$

i.e. every exponent is divisible by k. The proof is identical with "2" replaced by "k": if $a_i = kb_i$ then $n = \left(\prod p_i^{b_i}\right)^k$, and conversely squaring becomes raising-to-the-k and uniqueness does the rest.

- Perfect **square**: all exponents divisible by 2.
- Perfect **cube**: all exponents divisible by 3. (E.g. $216 = 2^3 \cdot 3^3 = 6^3$.)
- Perfect **sixth power**: all exponents divisible by 6 — these are exactly the numbers that are *both* squares and cubes (since $\mathrm{lcm}(2,3) = 6$). (E.g. $64 = 2^6 = 8^2 = 4^3$.)

1.9 Remark — Square-Free Numbers Are the Opposite Extreme

A number is **square-free** when *every* exponent equals 1 — the maximal violation of "all exponents even." Such numbers (like $105 = 3 \cdot 5 \cdot 7$ from Problem 1) are divisible by no perfect square other than 1.

Every positive integer factors *uniquely* as

$$n \;=\; s \cdot t^2,$$

where s is square-free (its **square-free part**) and t^2 is the largest square dividing n. The number n is a perfect square exactly when its square-free part is $s = 1$. For example $72 = 2 \cdot 36 = 2 \cdot 6^2$, so 72 has square-free part $2 \neq 1$ — not a square — whereas $36 = 1 \cdot 6^2$ has square-free part 1. This $s \cdot t^2$ split is also exactly what you do to "simplify a radical": $\sqrt{72} = \sqrt{2 \cdot 36} = 6\sqrt{2}$.

1.10 Remark — How Many Perfect Squares Are There?

Perfect squares thin out as numbers grow. The perfect squares up to N are $1^2, 2^2, \ldots, \lfloor \sqrt{N} \rfloor^2$, so there are exactly

$$\lfloor \sqrt{N} \rfloor$$

of them. Up to 100 there are 10 squares; up to $10\,000$ there are 100. The *density* of squares near N is about $\frac{1}{2\sqrt{N}} \to 0$ — squares become arbitrarily rare. This is the same $\sqrt{N}$ quantity that governs the sieve cutoff (Problems 2 and 3), reappearing here because squaring and square-rooting are the operations tying "size N" to "exponent structure."